

Literature Review: Robotic Arc Welding with the UR5e

MCTR911 Robotics Programming, Winter 2026 | Milestone 01 | Team: [team name / members]

1. Background

Arc welding joins metal parts along a seam using an electric arc, and it is one of the oldest applications of industrial manipulators. A robot suits the job because it can hold the torch's travel speed, angle and distance to the work constant along the whole seam, which is hard for a human to do over a full shift. Even so, many workshops still program welding robots by "teach and playback" (a person guides the arm through the seam once and the robot replays it) or by offline programming. Both adapt poorly when the workpiece or its position changes. Collaborative robots (cobots) are changing this. They are designed to work beside people without safety cages, and recent reviews argue they suit welding because they take over repetitive, strenuous work while skilled welders stay in charge of weld parameters, inspection and quality. The same reviews list AI integration, digital twins, open-source programming and better human-machine interfaces as open research needs.

2. Cobot Welding in Industry and Research

Industry. Welding systems built on Universal Robots (UR) arms first appeared at the FABTECH trade show in 2017, and by 2021 UR described welding and cutting as one of its fastest-growing markets, with partners selling MIG, TIG and plasma solutions. Ready-made products such as the Vectis cobot welding tool are built around a UR10e or UR5e mounted on a portable, electrically isolated cart. UR's own setup guidance for older CB3-series robots notes that a basic welding setup has no tracking or search, so the part must be positioned accurately by fixtures. In other words, the robot is only as good as its path and its part placement.

Research. Researchers address that limitation with sensing. One approach scans the workpiece into a point cloud, extracts the weld seam automatically and generates a smooth 6-DOF welding path (three position coordinates plus three torch-orientation angles), tested on several joint types and poses. Another approach mounts an RGB-D camera next to the torch on a UR3 arm to detect the groove and generate the welding trajectory. Both treat the weld as a 6-DOF path that the arm must track accurately, which is exactly the trajectory-generation and control problem this course covers.

3. The UR5e

The UR5e is a six-joint collaborative arm. Its 5 kg payload limits what can be mounted on the wrist, so the torch, cable holder and any camera must all fit within it. Its 850 mm reach defines the size of the work cell. Its pose repeatability of ± 0.03 mm (ISO 9283) matters for following a seam precisely. The controller updates at 500 Hz, which gives the smooth, closely timed joint commands that trajectory tracking needs, and the tool moves at about 1 m/s, far above typical welding travel speeds. An integrated force/torque sensor also enables hand-guided teaching and contact-based behaviors. Its standard six-revolute serial structure makes it a clean subject for frame assignment, DH parameters, forward and inverse kinematics, Jacobians, and trajectory and control design.

4. Why We Chose the UR5e for Welding

- **It fits the brief.** Six degrees of freedom is inside the required 4-7 range. Welding needs full pose control, because the torch's angle relative to the seam matters as much as its position, and a six-joint arm can command both. The five-DoF hobby arms in our candidate list cannot hold an arbitrary torch orientation.
- **It is credible as an industrial application.** UR arms appear in commercial welding cells and in welding research, so the application can be justified directly from the literature, which the project brief asks for.
- **It is ready for our toolchain.** A UR5e model with its visual meshes exists in MuJoCo Menagerie, and UR arms have ROS 2 support, which matches our ROS 2, MuJoCo and Python setup.
- **It is the right size.** Small workpieces sit well within its reach, and a light torch fits its payload. The larger UR10e would add reach and weight we do not need.

5. How Welding Maps to the Course Milestones

Welding exercises every part of the project in a natural way. In Milestone 2, the torch tip becomes the end-effector frame, and we derive the DH table and forward and inverse kinematics to place it on the seam. In Milestone 3, the weld requires a constant travel speed, so we use velocity kinematics (the Jacobian) to convert the desired tool speed into joint velocities, and we study how singularities disturb this. In Milestone 4, seam following is a task-space trajectory (straight lines, arcs and corners), while approach and retreat moves are joint-space trajectories. We then add motor-level control and compare the tracking error of the two trajectory types inside a simulated MuJoCo welding cell.